Introduction to Analytics

DATA SCIENCE

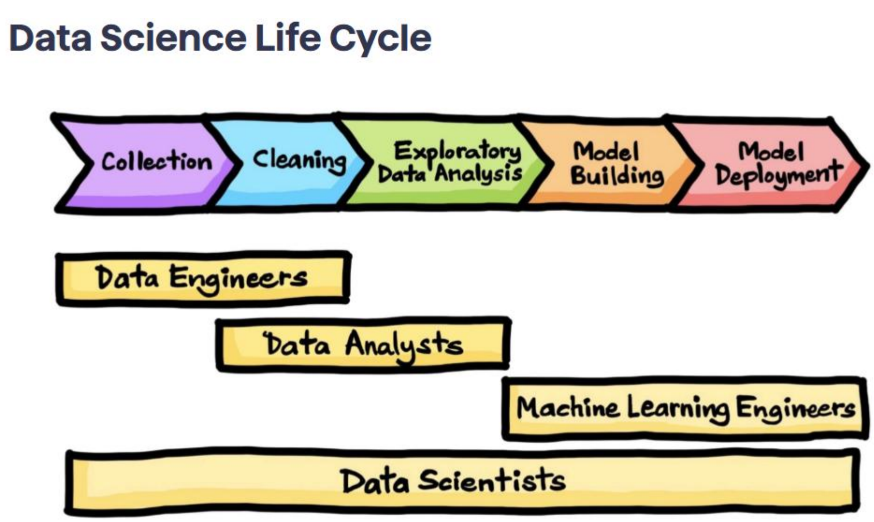
What is Data?

Data is a collection of facts, obtained as result of observations or experiments. It may consists of numbers, text or images. It is a source of information and knowledge.

WHAT IS DATA SCIENCE

- **Data Science** is an interdisciplinary field that uses **Statistics, Mathematics, programming (python/R), Machine Learning and domain expert knowledge** to **collect, process, analyze, and interpret** vast volume of data in order to **extract meaningful insights and support decision-making** and strategic planning.
  In simple words, Data science is turning raw data into useful information and predictions.

- The data science gives a systematic approach and structured framework for articulating the data problem as a question, deciding how to solve it and present the solution to stakeholders.



Components of Data Science

1. **Data Collection** – Gathering data from multiple sources (databases, APIs, IoT devices, etc.).
2. **Data Cleaning** – Removing errors, handling missing values, and ensuring data quality.
3. **Exploratory Data Analysis (EDA)** – Identifying patterns, trends, and relationships.
4. **Data Modeling** – Using machine learning and AI to make predictions or automate decisions.

5. **Data Visualization** – Presenting insights using charts, graphs, and dashboards.
6. **Deployment & Maintenance** – Implementing models into real-world applications and continuously improving them.

DATA SCIENCE PREREQUISTES

- **MACHINE LEARNING:** It is the back bone of data science. Solid grasp of ML is important with basic knowledge of statistics.
- **MODELING**: Modeling is a part of ML and involves identifying which algorithm is the most suitable to solve a problem and train the model.
- **STATISTICS:** It helps to extract more meaningful results.
- **PROGRAMMING**: Most used programming languages are Python and R. Python provides multiple libraries for data science and ML.
- **DATABASE**: A capable data scientist needs to understand how database works, how to manage them and how to extract data from them.

DATA SCIENCE ROLES

- **Data Engineer:** Builds and maintains pipelines that move, clean, and store data so others can use it.
- **Data Scientist:** leads the core role. Combines statistics, programming, and domain knowledge to build predictive models and extract actionable insights from data.
- **ML Ops Engineer:** They are responsible for putting the ML models prepared by traditional data scientists into production
- **Business Intelligence Analyst:** Focuses on business performance. Builds dashboards and reports that give executives a real-time view of company health.
- **Data Analyst:** The storyteller of data. Focuses on interpreting existing data to answer business questions and track performance metrics.
- **Data Product Manager:** This person is accountable for the success of a data product. They are responsible for defining, building, and managing data driven products; ensuring that data assets, pipelines, dashboards, ML models and analytics platforms deliver real business value. They strategize the most effective ways to execute the production process.

  All these roles are interconnected- Data Engineers feed clean data to Data Scientist, who build models that ML engineers deploy, while Analyst and BI teams report on outcomes, all governed by Architects and led by Managers.

DATA SCIENCE TOOLS

- Data Ananlysis: SAS, Jupyter, R studio, Matlab, Excel
- Data Warehousing: Informatica, Talend, AWS Redshift
- Data Visualisation: Tableau. Spark Mlib, Power BI, Mahout, Azure ML studio

Data Analytics

Analytics is the process of examining data to uncover useful insights, patterns, and trends that help in decision-making. It involves the use of various techniques, tools, and technologies to analyze raw data and transform it into meaningful information.

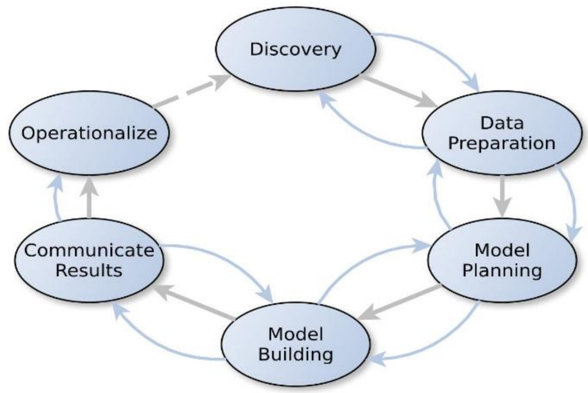
Types of Analytics

1. **Descriptive Analytics** – Answers **"What happened?"**
   - Summarizes historical data to identify trends and patterns.
   - Example: Monthly sales reports, customer demographics. Analyzing last month's sales data to see which products sold the most.
2. **Diagnostic Analytics** – Answers **"Why did it happen?"**
   - Investigates data to understand the causes of past events.
   - Example: Investigating why a specific product's sales dropped by analyzing customer reviews and competitor pricing.
3. **Predictive Analytics** – Answers **"What is likely to happen?"**
   - Uses statistical models and machine learning to forecast future outcomes.
   - Example: Predicting sales growth based on past trends. Forecasting next month's demand for a product based on past sales trends and seasonal patterns.
4. **Prescriptive Analytics** – Answers **"What should be done?"**
   - Provides actionable recommendations based on data-driven insights.
   - Example: Optimizing pricing strategies to maximize profit. Recommending discounts or promotions to boost sales for slow-moving products

Life Cycle of Data Analytics

The Data analytics lifecycle was designed to address Big Data problems and data science projects.

It encompasses the process of producing, collecting, processing, using, and analyzing data in order to meet corporate objectives. It outlines the step-by-step process used to extract actionable insights from data. Each phase of this cycle ensures that raw data is transformed into meaningful insights for effective decision-making.



Data Analytics Life Cycle – Healthcare Industry

The **Data Analytics Life Cycle** consists of several stages that guide the process of extracting insights from data.

Phase 1: Discovery

This is the **initial phase**, where the business problem is defined and understood.

- Define the **business problem** or objective to be solved.
- Identify the **scope of the data analytics project**.
- Determine relevant **data sources** (internal and external) and frame **initial Hypothesis**.
- Collaborate with stakeholders to **clarify objectives** and **requirements**, before touching the data.
- Assess available **resources**, including data, tools (descriptive statistics, data visualization) and team expertise.
- Example: exploring health drink data to identify trends and patterns.

Phase 2: Data Preparation (most time consuming phase)

This phase focuses on collecting and cleaning data to make it usable, as poor data leads to inaccurate results.

- **Collect data** from various sources like databases, APIs, and spreadsheets.
- Task include **ETL (extract, transform, load),** data cleaning, preprocessing, feature engineering.
- **Clean the data** by handling missing values, removing duplicates, and correcting inconsistencies.
- **Transform and format data** to make it suitable for analysis.
- Ensure data is **standardized and normalized** where necessary, to maintain data quality and consistency.
- Tools used are **correlation, covariance, and inferential statistics.**
- **Example:** cleaning, and transforming raw stock data for investment analysis.

Phase 3: Model Planning

Here, analyst decide how to **analyze the data**.

- The task include defining problem statements and objectives, selecting relevant **variables** (features) and choosing suitable algorithm (e.g.regression, clustering, classification) and techniques based on type of problem.
- The consideration include model complexity, interpretability and scalability.
- Develop a **roadmap** for the modeling phase, outlining how the data will be used.
- Create an initial **hypothesis** about how the model will behave with the chosen data.
- Use **exploratory data analysis (EDA)** techniques like correlation matrices and scatter plots to understand relationships in the data.
- Example: planning a ML model for customer churn prediction.

Phase 4: Model Building

At this phase the model is actually created and tested

- **Build models** using the selected algorithms and the prepared dataset. Its purpose is to develop and train predictive models.
- The task include splitting data into **training and testing data,** building and training models, fine-tuning model parameters. Its importance is for validation and for evaluation metrics.
- **Train models** by feeding them the training dataset to allow them to learn patterns.
- Use a **test dataset** to validate model performance and avoid overfitting.
- **Iterate and refine** the model by adjusting parameters for better accuracy.
- Use cross-validation techniques like **K-fold validation** to ensure robustness.
- Example: building a neural network for image recognition.

PHASE 5: Communicate Results

- To **document and communicate** finding.
- The task include **creating reports, dashboards, visualizations** and documenting insights and recommendations.
- Emphasize on **clear and effective communication** to stake holders
- Example: presenting data analysis results to company executives.

PHASE 6: Operationalize

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- **Deploy the model** into production environments, integrating it into business processes.
- Automate tasks like **decision-making** or predictions based on the model's insights.
- Emphasize on **quality assurance and improvement.** Continuously **monitor model performance** to ensure it remains accurate and relevant as new data becomes available.
- Update or retrain the model as needed to accommodate **changing data trends**.
- Document the deployment and maintenance processes to ensure **long-term usability**.
- Examples: deploying a fraud detection model in banking systems.

E-commerce example:

[Problem]

“Sales are decreasing. Why?”

[Collect Data]

Website visit, orders, customer behaviour

[Clean Data]

Remove errors, fill missing values

[Explore]

Find patterns : [High cart abandonment]

[Analyze]

Identify cause : [High shipping cost]

[Visualize]

Charts showing drop -off at checkout

[Act]

Introduce free shipping over Rs. 999

[Monitor]

Sales increase? If yes: refine further.

If no: re-analyze the problem

This cycle is not one-time, it keeps repeating. Every new decision creates new data, which leads to better insights over time.

Quality Assurance (QA) and Documentation

QA ensures the model and result are accurate and reliable.

- The model is rigorously tested for accuracy, reliability, bias, and edge cases.
- Ensuring the model is free from errors and performs correctly before any real- world deployment.
- Evaluating model performance using accuracy, precision, recall, and F1-score metrics.
- Recording (documenting) all processes, methodologies, and assumptions and results for future reference and improvements. This ensures reproducibility, transparency, smooth knowledge transfer to other teams.

Management Approval and Installation

- Getting approval from stakeholders before deployment. Presenting the results to stakeholders for review, validation and feedback. It includes ROI justification, strategic alignment, risk assessment (compliance, data privacy, operational risks)

  (ROI is the process of proving an investment is worthwhile by comparing expected gains against costs.)

- Get formal sign – off to move from "Experiment" to "production initiative".
- **Installation: Deployment of** the final approved system into a live environment- integrated with existing systems, API's, infrastructure, with monitoring tool setup.

  It also includes setting up monitoring, logging, access controls, and a rollback plan if something goes wrong.

Acceptance & Operations

Ensuring the system is fully functional and used effectively.

- **User acceptance testing (UAT)**: Stake holders test that outputs are correct, usable and meet requirements.
- End-users learn how to interpret model outputs or use dashboards.
- Track model performance, drift, data quality and system uptime.
- Operational procedures: SLAs, user training, feedback loops are established for ongoing use.
- Regular updates to data pipelines, models (retraining), and documentation as condition change.

Intelligent Data Analysis

- The final, continuous phase where **advanced AL/ML techniques** are applied for automated insights, pattern recognition and adaptive learning- making system smarter over time.

- Continuous monitoring and improvement of the model using new data insights. Regularly updating the model based on new data to improve accuracy and effectiveness.
- **Visualize results** through charts, graphs, dashboards, and other visuals to make insights understandable.
- Present key findings to **stakeholders** in a concise, actionable format.
- **Summarize the insights** derived from the data and explain how they align with business goals.
- Provide **recommendations** on actions based on the data analysis.
- Prepare a comprehensive **report** that includes all aspects of the data analysis process, findings, and conclusions.

LIFE CYCLE OF DATA ANALYSIS IN HOSPITAL

1. Data Collection – Gathering relevant data from various sources.

- **Example:** A hospital collects patient records, lab test results, treatment history, and doctor's notes.

2. **Data Cleaning** – Removing errors, missing values and inconsistencies.

- **Example:** Correcting missing or duplicate patient IDs, ensuring all records have valid timestamps.

3. **Data Exploration & Analysis** – Identifying patterns and trends and relationship in data.

- **Example:** Analyzing patient admission trends to identify peak hours and common illnesses.

4. Data Modeling – Applying statistical models or machine learning techniques for predictions.

- **Example:** Using predictive analytics to identify patients at high risk of developing diabetes based on historical health records.

5. Data Visualization & Interpretation – Presenting insights through dashboard, reports and charts for better understanding.

- **Example:** Creating a Power BI dashboard showing disease prevalence by age group and region.

6. Decision-Making & Action – Using insights to implement data driven informed business decisions.

- **Example:** Hospital management allocates more resources (beds, staff, and medical equipment) to departments treating high-risk patients.

Importance of Analytics

- Enhances decision-making.
- Improves business efficiency and competitiveness.
- Helps identify new opportunities and mitigate risks.
- Provides personalized experiences for customers.